Chemistry - Undergraduate Level Assessment 25 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Which of the following lists the hydrides of group-14 elements in order of thermal stability, from lowest to highest?
   (a) PbH 4 < SnH 4 < GeH 4 < SiH 4 < CH 4
   (b) PbH 4 < SnH 4 < CH 4 < GeH 4 < SiH 4
   (c) CH 4 < SiH 4 < GeH 4 < SnH 4 < PbH 4
   (d) CH 4 < PbH 4 < GeH 4 < SnH 4 < SiH 4

2. Calculate the Larmor frequency for a proton in a magnetic field of 1 T.
   (a) 23.56 GHz
   (b) 42.58 MHz
   (c) 74.34 kHz
   (d) 13.93 MHz

3. Of the following ionic substances, which has the greatest lattice enthalpy?
   (a) MgO
   (b) MgS
   (c) NaF
   (d) NaCl

4. A buffer is made from equal concentrations of a weak acid and its conjugate base. Doubling the volume of the buffer solution by adding water has what effect on its pH?
   (a) It has little effect.
   (b) It significantly increases the pH.
   (c) It significantly decreases the pH.
   (d) It changes the pH asymptotically to the pKa of the acid.

5. The strongest base in liquid ammonia is
   (a) NH 3
   (b) NH 2-
   (c) NH 4+
   (d) N_2H_4

True / False (10 marks)

Answer True or False

6. True or False: The 13 C spectrum of 2-methylpentane has lines with three distinct chemical shifts due to its unique branching.
7. True or False: Barium hydroxide ($Ba(OH)2$) can be directly converted to barium metal (Ba) without any intermediate steps.
8. True or False: The equilibrium polarization of 13 C nuclei is higher at lower magnetic field strengths, such as 10.0 T.
9. True or False: The atomic radii of the lanthanide elements decrease across the period from La to Lu due to increasing effective nuclear charge.
10. True or False: In the balanced equation, the ratio of MnO 4- to I- is 6:5.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss how the chemical environment of CH 3 Cl influences its proton NMR resonance frequency in a 12.0 T magnetic field, and explain the significance of this observation in spectroscopy.
12. Discuss the factors influencing the ratio of line intensities in the EPR spectrum of the t-Bu radical (CH 3)3 C$\bullet$, and analyze how these factors contribute to the observed intensity distribution.

End of Paper